Message framing:

Client messages will start with a 32 bit unsigned integer representing the size of the file being sent, followed by a binary byte array of the contents of the file.

Server messages will start with a 32 bit unsigned integer return code (0, 1, 2, 3), then a 32 bit unsigned integer representing the size of the URL character array that will be sent, followed by the character array of the URL.

How the end of a message is detected

The programs will stop reading messages once they have read the number of bytes that they were told to read by the incoming message. Each message has a fixed format at the start, which includes a size variable that conveys how many more bits to read to capture the whole message.

What happens on early disconnect

On early disconnect, the server will timeout, and a human readable character array will be sent to the client. This is done to prevent the server from hanging, as typically it must wait for the client to respond. Instead of trapping resources, after a set amount of time, the server can assume that a client has disconnected early, and close the connection.

AI use:

We searched how to read and send text files in tcp infrastructure and google overview provided us with the code to do it which was then modified. We also had to research how to set up the zebra crossing library.

Sources Used

https://chat.deepseek.com/share/wkq834cnjnu6y8z2bf